CAPSTONE ENGINEERING CAPSTONE

LAMINA STUDIOS

- Website Deployment
- Project Deliverables: Portfolio, Demo Video, and GitHub Repository
- Automated CI/CD Pipeline
- Dockerized Application Running on Minikube Pods
- Infrastructure as Code (IaC) with Terraform
- Monitoring with Prometheus and Grafana

WHAT I DO

- Triggered on push or pull request to main branch
- Checkout code from GitHub repository
- Setup Node.js 20 & install frontend dependencies
- Build React frontend (Vite v8)
- Setup Docker Buildx for container builds
- Docker Login using GitHub secrets
- Build & push backend image: shirahoshiii/backend:latest
- Build & push frontend image: shirahoshiii/frontend:latest
- Deploy to Kubernetes: apply namespace, deployments & services
- Deployment summary: verify Docker images and namespace

GITHUB ACTIONS

TECHOLOGIES

Website Deployment (Local Host)
Minikube
DOCKER CONTAINER
IaC
Prometheus
Garafan
Github Actions

THANK YOU

LAMINA STUDIOS